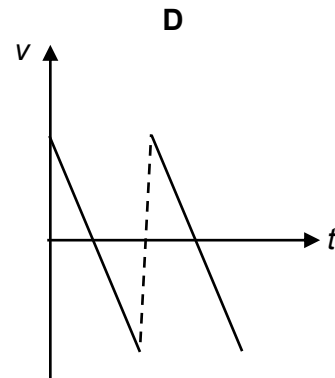
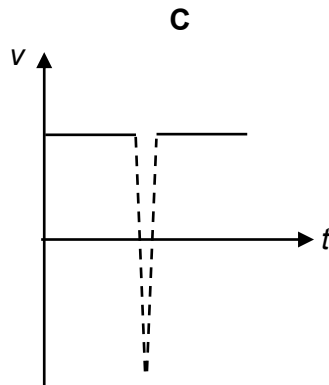
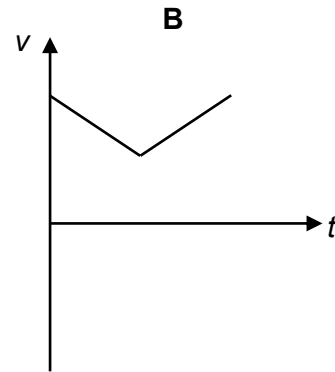
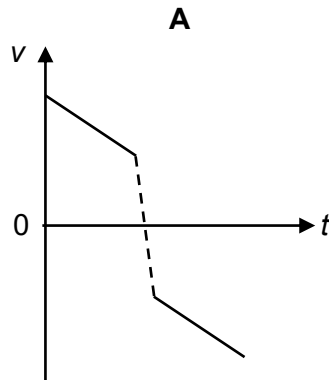


smaller) and hence causes systematic errors

- 2 A tennis ball is thrown vertically upward. It hits the ceiling before it falls down. Assuming the effect of air resistance is negligible, which graph best represents the variation of velocity v with time t before and after hitting the ceiling?



Ans: A

Before hitting and after hitting ceiling, the ball is under free fall, with a constant downward acceleration of 9.81 m s^{-2} , where straight downward diagonal lines are expected. During the collision with the ceiling, the downward acceleration is greater than 9.81 m s^{-2} as the ceiling, in addition to the weight of the ball, exerts a downward force $\rightarrow$ hence a more negative gradient.

- 3 An object is projected with velocity 40 m s^{-1} at an angle of 60° to the horizontal. Air resistance is